COURIER SORTING AUTOMATED MACHINE

PLC I/O • WIRING PREPARATION • STATION-WISE CABLE ROUTING • PNEUMATICS

1. Purpose and Work Sequence

Use this reference to prepare wires, route cables by station, label both ends, and complete terminations systematically. The PLC addresses below are retained from the supplied notes; verify every terminal against the actual PLC, sensor, relay and valve documentation before energizing.

Recommended sequence before making connections

Step	Preparation activity
01	Study the electrical drawings, PLC I/O list, terminal plan and pneumatic schematic. Confirm device tags, voltage, cable type and wire numbers.
02	Measure each route physically along the intended trunking/conduit path, including vertical drops, bends, service loops and movement allowance for cylinders or moving parts.
03	Cut each wire to the measured route length plus a practical termination/service allowance at both ends. Avoid cutting all wires to one common length.
04	Strip insulation to the terminal manufacturer's specified length; use the correct ferrule/lug and crimp tool for stranded wires. Do not nick copper strands.
05	Label both ends before pulling wires. Use consistent tags such as ST1-LPE, ST1-RPE, ST1-RSV, ST1-LSV and corresponding PLC I/O addresses.
06	Bundle and route wiring by station and function. Keep 24 VDC control/sensor wiring separate from AC mains, VFD motor output cables and other noise sources; cross at right angles where practical.
07	Route pneumatic tubing separately from electrical wiring where practical. Keep tubing away from sharp edges, hot surfaces and pinch points; allow cylinder movement and secure tubing against abrasion.
08	Terminate one circuit at a time, check ferrules and terminal tightness, then perform point-to-point continuity and short-circuit checks before applying power.

2. Station-Wise Routing Plan

Use separate labelled bundles/trunking branches for the entrance, Station 1 and Station 2. Route each device cable from the field device to the station junction/terminal strip, then to the PLC panel terminals. This keeps fault-finding clear and reduces rework.

Area	Circuit group	Recommended route / connection reference
Entrance	Camera Ethernet	Camera → Ethernet switch/module → PLC/network port; use suitable Ethernet cable and keep it away from VFD output wiring.
Entrance	Photoelectric sensors	Sensor → entrance terminal strip → PLC input X3 (per supplied notes); brown to +24 VDC and blue to 0 VDC for the noted 3-wire PNP sensor; black output to X3.
Entrance	START / STOP	Push buttons → labelled terminal strip → PLC inputs X0 / X1; confirm the input common and NO/NC contact scheme from the actual hardware.
Entrance	Emergency_OK	Safety/status circuit → relay/contact as designed → PLC input X2. Do not use ordinary PLC logic as the sole emergency-stop safety function.
Station 1	Sensors and reset PBs	St1_Left_PE X4; St1_Right_PE X5; Left_ResetPB X6; Right_ResetPB X7. Route as a labelled Station 1 input bundle.
Station 1	Indicator lamps	St1_Left_Lamp Y5 via relay; right lamp Y6 via relay (confirm right-lamp tag before labelling). Route output wiring through the designated terminal strip.
Station 1	Cylinder sensors	Right-side cylinder reed switch X15; left-side cylinder reed switch X16. Route sensor leads with enough slack for full cylinder travel; secure against rubbing.
Station 1	Solenoid valves	Right-side cylinder solenoid Y1; left-side cylinder solenoid Y2 per supplied notes. PLC output → interposing relay/driver if required by the valve coil current and output rating → valve coil. Verify coil voltage, suppression and output type.
Station 1	Barcode scanner	Scanner → PLC/network interface as designed; address/protocol not specified in source notes. Route Ethernet/data cable separately from power wiring.
Station 2	Sensors and reset PBs	St2_Left_PE X10; St2_Right_PE X11; Left_ResetPB X12; Right_ResetPB X13. Route as a labelled Station 2 input bundle.
Station 2	Indicator lamps	St2_Left_Lamp Y7 via relay; St2_Right_Lamp Y10 via relay. Route output wiring through the designated terminal strip.
Station 2	Cylinder sensors	Right-side cylinder reed switch X17; left-side cylinder reed switch X20. Leave movement allowance and secure cable/tubing.
Station 2	Solenoid valves	Right-side cylinder solenoid Y2; left-side cylinder solenoid Y3 per supplied notes. Verify the Y2 address conflict noted in Section 5 before wiring.
Station 2	Barcode scanner	Scanner → PLC/network interface as designed; address/protocol not specified in source notes. Route Ethernet/data cable separately from power wiring.

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WIRING CHECKLIST • PNEUMATIC VALVES • COMMISSIONING

3. Pneumatic and Solenoid Connection Sequence

Item	Connection / work instruction
Air supply	Compressor/plant air → lockable isolation valve → filter/regulator (and lubricator only if required by the valve/cylinder manufacturer) → manifold/supply port.
Directional control valve	Identify the valve ports from its printed markings and datasheet. For a typical 5/2 valve: P/1 = pressure supply; A/2 and B/4 = cylinder ports; exhaust ports 3 and 5 go to silencers or safe exhaust routing. Confirm the exact valve symbol/port numbering.
Cylinder tubing	Connect A and B to the two cylinder ports according to the pneumatic schematic. Swap A/B only if the intended extend/retract action requires it and the design permits.
Electrical coil	PLC output → interposing relay/driver if needed → solenoid coil; provide the specified supply/common return. Check coil voltage and current against PLC/relay ratings.
Feedback	Cylinder reed switch → corresponding PLC input (Station 1 X15/X16; Station 2 X17/X20). Confirm sensor supply, output type and input common.
Safety and test	Isolate and depressurize before tubing changes. Secure tubing, check leaks at low pressure, keep people clear of the cylinder stroke, and test one actuator at a time.

4. Wire Cutting, Identification and Routing Checklist

■ Route measured and recorded for each cable/wire

■ Wire cut to individual length with termination/service allowance

■ Both ends labelled with device tag and PLC address

■ Ferrules/lugs crimped with correct tool and strip length

■ Station 1 and Station 2 bundles separated and secured

■ Sensor, data, control, mains and VFD motor wiring segregated as appropriate

■ Pneumatic tubes labelled, secured and clear of moving parts

■ Terminal numbers match the drawings and I/O list

■ Continuity, polarity, short-circuit and earth checks completed

■ Power-up and actuator tests completed under an approved safe procedure

5. Items to Confirm Before Energizing

• **Duplicate output address:** Y2 is listed for both the Station 1 left cylinder solenoid and Station 2 right cylinder solenoid. Confirm PLC hardware mapping and program assignment; one physical output cannot independently control two intended actuators unless specifically designed that way.

• **Station 1 right lamp tag:** Source notes use “St1_Left_Lamp” at Y6. Confirm whether the tag should be St1_Right_Lamp.

• **Push-button input logic:** Confirm PLC input type and push-button contact arrangement; do not assume both 0 V and 24 V switching schemes apply.

• **Emergency stop:** Use a properly designed, appropriately rated safety circuit. The PLC X2 status input is not a substitute for the safety function.

• **Solenoid output protection:** Check output current rating, coil voltage, interposing relay/driver requirement and manufacturer-recommended flyback/surge suppression.

• **PNP sensor wiring:** Confirm model-specific pinout and PLC input compatibility before applying power.

Document basis: original Courier Sorting I/O notes supplied by the user. Routing and preparation steps are practical recommendations; final connections must follow the actual electrical drawings, device datasheets and approved safety design.